WILLIAM ONG

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LinkedIn ◇ Github

SUMMARY

Software Engineer with 3+ years of experience building high-throughput, distributed systems in payments, ride-hailing, and logistics. Strong in Go and JVM-based backend architecture, experienced in building high-throughput, distributed systems in payments, ride-hailing, and logistics, including financial-grade workflows, payment APIs, and event-driven architectures.

EXPERIENCE

Senior Software Engineer (L3) in Gojek (Allocation Platform) Oct 2023 - Present
Gojek is a **leading super app** in Indonesia (from ride-hailing, food delivery and more). *Jakarta, Indonesia*

- Scales and develop Constraint Enricher, a high-throughput Go-based rule engine serving as the source of truth for driver allocation filters, managing 100k RPM peaks with P99 under 60ms via an optimized periodic-refresh caching strategy.
- Unified cross-team driver eligibility logic into a Kubernetes-selector–inspired rule engine built in Go, improving correctness and eliminating config inconsistencies, reducing launch issues from 80% → near-zero in 6 months.
- Build Aggregator, a Go based real-time aggregator processing 80k+ records/s, leveraging S2ID-based spatial indexing and HyperLogLog (HLL) sketches to provide sub-second supply-demand signals for allocation logic while optimizing ScyllaDB for high-throughput, conflict-free writes.
- Built internal platforms Stargate end-to-end (Vue 3 + Express) enabling real-time driver observability dashboard, and chat-based AI Agent for faster knowledge retrieval and operational queries.
- Implemented refresh-ahead caching in config service, reducing P99 latency by 70% (75ms to 15ms) for services handling 1,000 RPS.

Software Engineer at Shopee (Payment Processing) Aug 2022 - Oct 2023
Shopee is the **largest e-Commerce application** in Southeast Asia. *Jakarta, Indonesia*

- Received Value Star Award (We Commit) for exemplifying strong ownership in leading high-impact initiatives.
- Automated payment transaction reconciliation process, reducing batch processing time from 1 day → 15 minutes for 2.5-3 million transactions.
- Re-architected the payment callback (webhook) system using event-driven architecture with Kafka, eliminating existing single point of failure and scaling from 100 to 200 callbacks/sec.
- Led incremental migration of core payment callback system from Java to Go, designing shadow traffic validation, and phased rollout strategies to ensure correctness and zero downtime across multiple banking partners.

EDUCATION

Bachelor of Computer Science, Bandung Institute of Technology (ITB) 2018 - 2022
Graduated with **3.95/4.0 GPA** (transcript). One of the highest in the entire class of 2022 graduate.

CERTIFICATIONS / AWARDS / LEADERSHIP

Value Star Awardee (2023). Recognized for demonstrating ownership and leadership by examplifying "We Commit" values in Shopee (certificate)

Speaker, ICAICTA (2022). Presented research on Generative Opinion Triplet Extraction Using Pretrained Language Model in the Sentiment Analysis & Information Validation track. (event reference)

Head of Data Competition Events (2021). Act as the head division on Datavidia, one of the data science competitions held by ITB during Arkavidia. (certificate), (event reference)

SKILLS

Languages	Go, Python, Java, TypeScript
Infrastructure	Kubernetes, Kafka, Redis, MySQL, PostgreSQL, Terraform, GitLab CI, Prometheus, Grafana